Smart Parking Management System
Final PostgreSQL Database Schema & Additional SRS Features

New Functional Requirements

FR26. Incident Reporting
Users can report blocked slots, damaged slots, wrong parking, payment issues, QR issues and attach descriptions/images. Admins can review, respond and resolve incidents.

FR27. Slot Reassignment
Admins can reassign booked parking slots. Previous QR codes become invalid and new QR codes are generated. All changes are logged for auditing.

FR28. Advanced Analytics
Dashboard includes peak hours, average parking duration, most/least used lots, revenue by vehicle type, booking trends, occupancy, cancellations, repeat users and incident statistics.

users

id (UUID) PK, full_name, email UNIQUE, password_hash, phone, profile_image, role(USER/ADMIN), is_verified, created_at, updated_at

vehicles

id PK, user_id FK, vehicle_number UNIQUE, vehicle_type, vehicle_model, vehicle_color, is_default, created_at, updated_at

parking_lots

id PK, name, address, latitude, longitude, total_slots, available_slots, description, created_at, updated_at

parking_slots

id PK, lot_id FK, slot_number, slot_type, status, hourly_price, floor, section, is_active, created_at, updated_at

bookings

id PK, user_id FK, vehicle_id FK, slot_id FK, booking_type, booking_status, booking_start, booking_end, entry_time, exit_time, duration_minutes, total_amount, qr_code, payment_status, created_at, updated_at

payments

id PK, booking_id FK, amount, tax, payment_method, payment_status, transaction_reference, paid_at

qr_codes

id PK, booking_id FK, qr_data, version, is_active, generated_at, expires_at

slot_reassignments

id PK, booking_id FK, old_slot_id FK, new_slot_id FK, changed_by FK(users), reason, changed_at

notifications

id PK, user_id FK, title, message, type, is_read, created_at

incident_reports

id PK, booking_id FK NULL, user_id FK, slot_id FK, incident_type, description, image_url, status, admin_response, resolved_by FK(users), created_at, resolved_at

admin_logs

id PK, admin_id FK, action, target_table, target_id, description, created_at

Relationships

Users -> Vehicles (1:M)
Users -> Bookings (1:M)
Vehicles -> Bookings (1:M)
Parking Lots -> Parking Slots (1:M)
Parking Slots -> Bookings (1:M over time)
Bookings -> Payments (1:1)
Bookings -> QR Codes (1:M for regenerated QR versions)
Bookings -> Slot Reassignments (1:M)
Users -> Notifications (1:M)
Users -> Incident Reports (1:M)

Admin Analytics

Live occupancy, Peak hours, Average parking duration, Most used lot, Least used lot, Revenue by vehicle type, Revenue by parking lot, Daily/Weekly/Monthly booking trends, Cancellation statistics, Incident statistics, Active users, Returning users.